

Mining Frequent Itemsets and Association Rules

□ Association rule mining

- Given two thresholds: $minsup$, $minconf$
- Find **all** of the rules, $X \rightarrow Y (s, c)$ such that $s \geq minsup$ and $c \geq minconf$

□ Let $minsup = 50\%$

- Freq. 1-itemsets: Beer: 3, Nuts: 3, Diaper: 4, Eggs: 3
- Freq. 2-itemsets: {Beer, Diaper}: 3

□ Let $minconf = 50\%$

- $Beer \rightarrow Diaper$ (60%, 100%)
- $Diaper \rightarrow Beer$ (60%, 75%)



TID	Items bought
1	Beer, Nuts, Diaper
2	Beer, Coffee, Diaper
3	Beer, Diaper, Eggs
4	Nuts, Eggs, Milk
5	Nuts, Coffee, Diaper, Eggs, Milk

□ Observations:

- Mining association rules and mining frequent patterns are very close problems
- Scalable methods are needed for mining large datasets

(Q: Are these all the rules satisfying the two conditions?)

Challenge: There Are Too Many Frequent Patterns!

- A long pattern contains a combinatorial number of sub-patterns
- How many frequent itemsets does the following TDB₁ contain (minsup = 1)?

□ TDB₁: T₁: {a₁, ..., a₅₀}; T₂: {a₁, ..., a₁₀₀}

□ Let's have a try

1-itemsets: {a₁}: 2, {a₂}: 2, ..., {a₅₀}: 2, {a₅₁}: 1, ..., {a₁₀₀}: 1,

2-itemsets: {a₁, a₂}: 2, ..., {a₁, a₅₀}: 2, {a₁, a₅₁}: 1 ..., ..., {a₉₉, a₁₀₀}: 1,

..., ..., ..., ...

99-itemsets: {a₁, a₂, ..., a₉₉}: 1, ..., {a₂, a₃, ..., a₁₀₀}: 1

100-itemset: {a₁, a₂, ..., a₁₀₀}: 1

- The total number of frequent itemsets:

$$\binom{100}{1} + \binom{100}{2} + \binom{100}{3} + \cdots + \binom{100}{100} = 2^{100} - 1$$

A too huge set for any one to compute or store!



Expressing Patterns in Compressed Form: Closed Patterns

- ❑ How to handle such a challenge?
- ❑ Solution 1: **Closed patterns**: A pattern (itemset) X is **closed** if X is *frequent*, and there exists *no super-pattern* $Y \supset X$, with the same support as X
 - ❑ Let Transaction DB TDB_1 : $T_1: \{a_1, \dots, a_{50}\}$; $T_2: \{a_1, \dots, a_{100}\}$
 - ❑ Suppose $minsup = 1$. How many closed patterns does TDB_1 contain?
 - ❑ Two: $P_1: \{\{a_1, \dots, a_{50}\}: 2\}$; $P_2: \{\{a_1, \dots, a_{100}\}: 1\}$
- ❑ **Closed pattern** is a **lossless compression** of frequent patterns
 - ❑ Reduces the # of patterns but does not lose the support information!
 - ❑ You will still be able to say: $\{\{a_2, \dots, a_{40}\}: 2\}$, $\{\{a_5, a_{51}\}: 1\}$

Expressing Patterns in Compressed Form: Max-Patterns

- ❑ Solution 2: **Max-patterns**: A pattern X is a **max-pattern** if X is frequent and there exists no frequent super-pattern $Y \supset X$
- ❑ Difference from close-patterns?
 - ❑ Do not care the real support of the sub-patterns of a max-pattern
 - ❑ Let Transaction DB TDB_1 : $T_1: \{a_1, \dots, a_{50}\}$; $T_2: \{a_1, \dots, a_{100}\}$
 - ❑ Suppose $minsup = 1$. How many max-patterns does TDB_1 contain?
 - ❑ One: $P: \{\{a_1, \dots, a_{100}\}: 1\}$
- ❑ **Max-pattern** is a **lossy compression**!
 - ❑ We only know $\{a_1, \dots, a_{40}\}$ is frequent
 - ❑ But we do not know the real support of $\{a_1, \dots, a_{40}\}$, ..., any more!
- ❑ Thus in many applications, mining close-patterns is more desirable than mining max-patterns